

DSA02-Computer Vision

Code of Conduct:

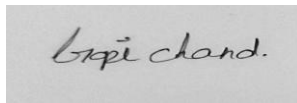
I G. Gopi Chand (192424384) (Name / Reg No) certify that this submission is my

original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student:

A rectangular box containing a handwritten signature in cursive script that reads "Gopi Chand.".

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation changes the brightness of pixels to improve image quality.

Uses:

- Improves contrast
- Brightens dark images
- Highlights important features
- Makes images easier to analyze

(b) Negative Transformation

Formula

$$s = L - 1 - r$$

Where

- $L = 256$ (8-bit image)
- r = original pixel value
- s = transformed pixel value

Suppose the pixel values are

50, 100, 150, 200 50, 100, 150, 200

Then

Original (r)	Negative (255-r)
50	205
100	155
Original (r)	Negative (255-r)
150	105
200	55

Answer:

205, 155, 105, 55 205, 155, 105, 55

2. Log Transformation

(a) Purpose

Log transformation increases the brightness of dark pixels more than bright pixels.

It is useful for

- Dark images

- Medical images
- Satellite images

Formula

$$s = c \log(1+r) \quad s = c \log(1+r)$$

(b) Calculation

Given

$$c = 1 \quad c = 1$$

Pixel values

0, 1, 3, 70, 1, 3, 7

Using

$$s = \log(1+r) \quad s = \log(1+r)$$

r	s = log(1+r)
0	0
1	0.693
3	1.386
7	2.079

Output

0, 0.693, 1.386, 2.079 0.693, 1.386, 2.079

3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization improves image contrast by spreading pixel intensities over the full range.

Advantages

- Better contrast
 - More visible details
 - Better segmentation
-

(b) Histogram Equalization

Given

Intensity	Frequency
0	2
1	2

Intensity	Frequency
2	4
3	8

Total pixels

$N=16$

Step 1: Cumulative Frequency

Intensity	Frequency	Cumulative
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Values

Since

$L=4$

Formula

$S_k = \text{CDFN}(L-1)S_k = NCDF(L-1)$

Intensity	CDF	Equalized
0	2	$0.375 \approx 0$
1	4	$0.75 \approx 1$

Intensity	CDF	Equalized
2	8	$1.5 \approx 2$
3	16	3

Equalized Intensities

0, 1, 2, 30,1,2,3

4. Spatial Filtering (Smoothing)

(a) Purpose

Smoothing removes noise by averaging neighboring pixels.

Applications

- Noise removal
 - Blur unwanted details
 - Improve segmentation
-

(b) 3×3 Averaging Filter

Suppose

[203040305060406070]203040305060406070

Average

$= \frac{20+30+40+30+50+60+40+60+70}{9} = \frac{400}{9} = 44.44 = 44$

Rounded

4444

New center pixel = 44

5. Spatial Filtering (Sharpening)

(a) Purpose

Sharpening highlights edges and fine details.

Applications

- Medical images
 - Satellite images
 - Text enhancement
-

(b) Laplacian Mask

Mask

$\begin{bmatrix} 0 & -10 & -14 & -10 & -10 \\ 0 & -10 & -14 & -10 & -10 \end{bmatrix}$

Suppose image

$\begin{bmatrix} 203 & 020 & 305 & 030 & 203 & 020 \\ 203 & 020 & 305 & 030 & 203 & 020 \end{bmatrix}$

Calculation

$$=(4 \times 50) - (30 + 30 + 30 + 30) = (4 \times 50) - (30 + 30 + 30 + 30) = 200 - 120 = 200 - 120 = 80 = 80$$

Output = 80

6. Frequency Domain Filtering

(a) Difference

Low Pass Filter	High Pass Filter
Keeps low frequencies	Keeps high frequencies
Removes noise	Enhances edges
Produces smooth image Produces sharp image	

(b) Example

Frequency components

[10, 8, 5, 2][10,8,5,2]

Retain first two

Output

[10, 8, 0, 0][10,8,0,0]

7. Thresholding

(a) Global Thresholding

A single threshold separates foreground from background.

Rule

- $\text{Pixel} \geq T \rightarrow 1$
 - $\text{Pixel} < T \rightarrow 0$
-

(b) Example

Pixels

40, 60, 70, 20, 8040,60,70,20,80

Threshold

$T=50$

Result

Pixel	Output
40	0
60	1
70	1
20	0

Pixel	Output
80	1

Final

0, 1, 1, 0, 10,1,1,0,1

8. Edge Detection

(a) Purpose

Edge detection finds object boundaries where intensity changes suddenly.

Applications

- Face detection
 - Medical imaging
 - Object recognition
-

(b) Sobel Horizontal

Mask

$[-1-2-1 \ 0 \ 0 \ 1 \ 2 \ 1] \ -101-202-101$

Suppose image

$[102030203040304050] \ 102030203040304050$

Calculation

$=(-10-40-30)+(30+80+50)=(-10-40-30)+(30+80+50)=-80+160=-80+160=80=80$

Gradient = 80

9. Region-Based Segmentation

(a) Region Growing

Starts from a seed pixel and adds neighboring pixels with similar intensity.

Advantages

- Produces connected regions
- Accurate segmentation

(b) Example

Pixels

48, 50, 52, 60, 49 48,50,52,60,49

Seed

5050

Threshold difference

55

Differences

Pixel	Difference	Included
48	2	Yes
50	0	Yes
52	2	Yes
60	10	No
49	1	Yes

Region

48, 49, 50, 52 48,49,50,52

10. Combined Enhancement & Segmentation

(a) Why Smoothing Before Segmentation?

Smoothing removes noise, making segmentation more accurate and reducing false edges.

(b) Example

Given image values

45, 48, 50, 49 45,48,50,49

Average (given)

48 48

Threshold

$T=48$ $T=48$

Rule

- $\geq 48 \rightarrow 1$
- $< 48 \rightarrow 0$

Pixel	Output
45	0
48	1
50	1
49	1

Final segmented output

0, 1, 1, 1 0,1,1,1

